Transformations, congruence, and similarity

ANSWER KEY

Core Concepts: Transformations, congruence, and similarity

1 In $\triangle ABC$ studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 84° ✓

b. 94°

c. 76°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 45^\circ) = 84^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 8

c. 6 ✓

d. 9

Substituting $x = 1$ into $y = 3x + 3$ gives $y = 3(1) + 3 = 6$.

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Procedural Fluency & Algebraic Manipulation

4 In $\triangle ABC$ studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 77° ✓

b. 87°

c. 69°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 52^\circ) = 77^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm ✓

c. 20 cm

d. 45 cm

Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 6

b. 11

c. 9 ✓

d. 12

Substituting $x = 2$ into $y = 3x + 3$ gives $y = 3(2) + 3 = 9$.

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 70° ✓

b. 80°

c. 62°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 59^\circ) = 70^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 9

b. 14

c. 12 ✓

d. 15

Substituting $x = 3$ into $y = 3x + 3$ gives $y = 3(3) + 3 = 12$.

Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 63° ✓

b. 73°

c. 55°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 66^\circ) = 63^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm ✓

c. 30 cm

d. 125 cm

Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 12

b. 17

c. 15 ✓

d. 18

Substituting $x = 4$ into $y = 3x + 3$ gives $y = 3(4) + 3 = 15$.

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